

IMO(43rd) – 2002

First Day

1. S is the set of all (h, k) with h, k non-negative integers such that $h + k < n$. Each element of S is colored red or blue, so that if (h, k) is red and $h' \leq h, k' \leq k$, then (h', k') is also red. A type 1 subset of S has n blue elements with different first member and a type 2 subset S has n blue element with different second member. Show that there are the same number of type 1 and type 2 subsets.
2. BC is a diameter of a circle with centre O . A is any point on the circle with $\angle AOC > 60^\circ$. EF is the chord which is the perpendicular bisector of AO . D is the midpoint of the minor arc AB . The line through O parallel to AD meets AC at J . Show that J is the incentre of the triangle CEF .
3. Find all pairs of integers $m > 2, n > 2$ such that there are infinitely many positive integers k for which $k^n + k^2 - 1$ divides $k^m + k - 1$.

Second Day

4. The positive divisors of integer $n > 1$ are $d_1 < d_2 < \dots < d_k$ so that $d_1 = 1, d_k = n$. Let $d = d_1 d_2 + d_2 d_3 + \dots + d_{k-1} d_k$. Show that $d < n^2$ and find all n for which d divides n^2 .
5. Find all real-valued functions on the reals such that :
 $(f(x) + f(y))(f(u) + f(v)) = f(xu - yv) + f(xv + yu)$ for all x, y, u, v .
6. $n > 2$ circles of radius 1 are drawn in the plane so that no line meets more than two of the circles. Their centres are $O_1, O_2, \dots, O_n$. Show that $\sum_{i < j} \frac{1}{O_i O_j} \leq (n-1) \frac{\pi}{4}$.

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